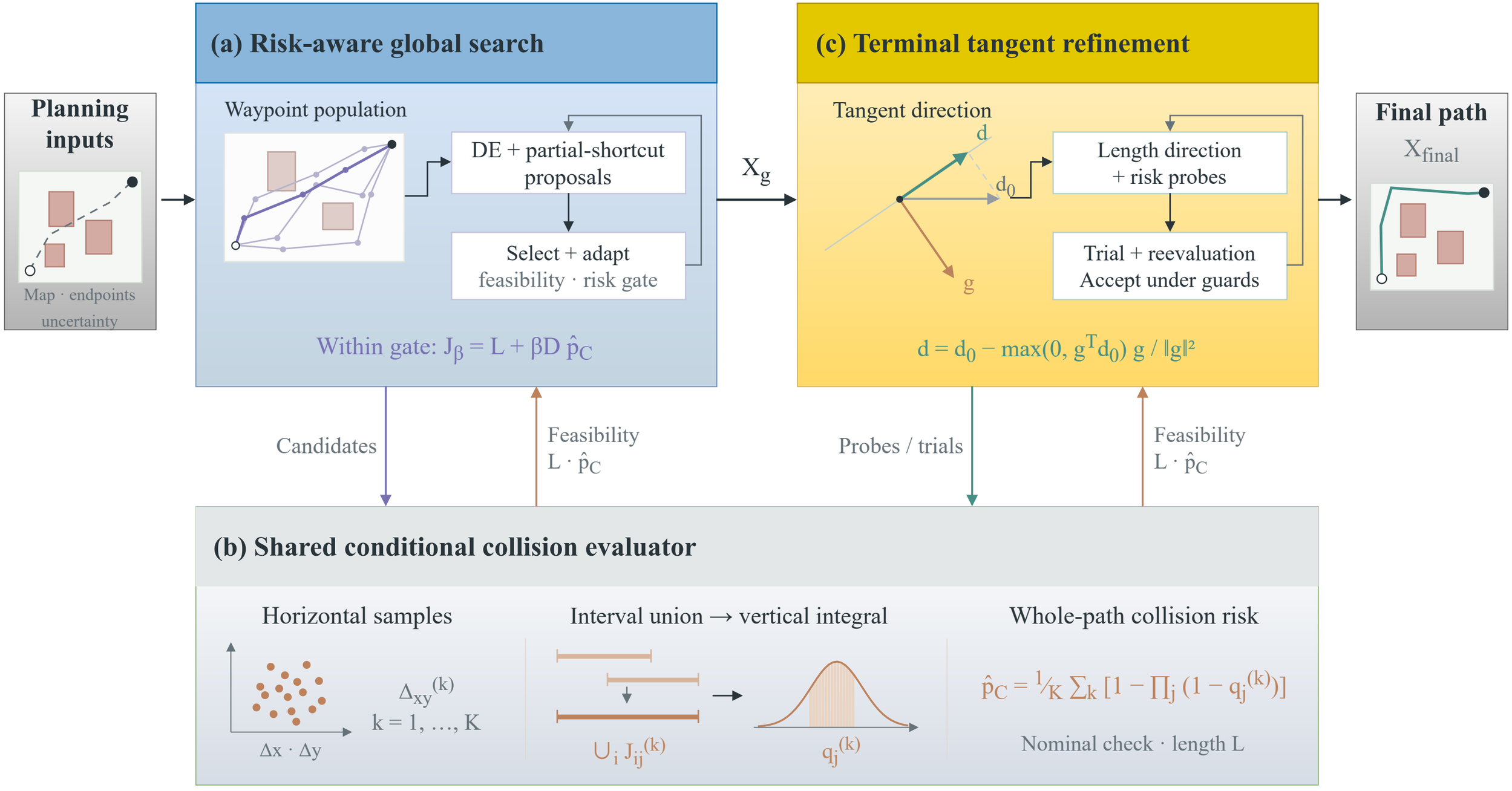

Planning inputs
Map · endpoints
uncertainty
(a) Risk-aware global search
Waypoint population
DE + partial-shortcut proposals
Select + adapt
feasibility · risk gate
Within gate: $J_\beta = L + \beta D\,\hat{p}_C$
X_g
(c) Terminal tangent refinement
Tangent direction
d
d_0
g
Length direction + risk probes
Trial + reevaluation
Accept under guards
$d = d_0 - \max(0, g^T d_0)\, g / \|g\|^2$
Final path
X_{final}
Candidates
Feasibility
$L \cdot \hat{p}_C$
Probes / trials
Feasibility
$L \cdot \hat{p}_C$
(b) Shared conditional collision evaluator
Horizontal samples
$\Delta x \cdot \Delta y$
$\Delta_{xy}^{(k)}$
$k = 1, \ldots, K$
Interval union → vertical integral
$\cup_i J_{ij}^{(k)}$
$q_j^{(k)}$
Whole-path collision risk
$\hat{p}_C = {}^1\!/_K \sum_k [1 - \prod_j (1 - q_j^{(k)})]$
Nominal check · length L